

Fine-Tuned BERT Based Multilingual Model for Named Entity Recognition in Native Indian Languages

1st G Bharathi Mohan

Computer Science and Engineering (AIE)

Amrita School of Computing Amrita Vishwa Vidyapeetham
Chennai, India

g_bharathimohan@ch.amrita.edu

3rd Elakkiya R

Computer Science and Engineering (CSE)

Birla Institute of Technology and Science

Dubai, United Arab Emirates

elakkiya@dubai.bits-pilani.ac.in

2nd R Prasanna Kumar

Computer Science and Engineering (AIE)

Amrita School of Computing Amrita Vishwa Vidyapeetham
Chennai, India

r_prasannakumar@ch.amrita.edu

4th Mukhtesh Venkata Sri Sai Pendem

Computer Science and Engineering (AIE)

Amrita School of Computing Amrita Vishwa Vidyapeetham
Chennai, India

mukhteshpendem31@gmail.com

Abstract—Accurate recognition of named entities is essential in real-world applications such as information extraction, document classification, sentiment analysis, and machine translation. The fine-tuned BERT model demonstrates exceptional performance in multilingual NER tasks by leveraging language-specific optimization, cross-lingual transfer, and context understanding. In contrast, other models applied to multilingual data may exhibit limitations. The research utilizes the WikiAnn dataset, including languages like Tamil, Malayalam, Bengali, and Marathi, to train the fine-tuned BERT model. The system achieved a precision of 95.3% and a recall of 94.8% for the Bengali language, outperforming the generic model used for Indian languages. It demonstrates superior accuracy and effectiveness in deciphering Bengali text. This research provides a robust solution for accurate named entity recognition across multiple languages, benefiting real-world applications.

Keywords—Named Entity Recognition, Bidirectional Encoder Representations from Transformers, Native Indian languages, WikiAnn dataset, Natural Language Processing, Machine Learning.

I. INTRODUCTION

Named Entity Recognition (NER) [1] is a subtask of Natural Language Processing (NLP) that includes detecting and extracting named entities from unstructured text data, such as persons, organizations, places, and other sorts of things. NER has a wide range of applications, including information retrieval, question-answering systems, and text summarization [2] [3]. NER is described as the automated recognition of proper names and their classification into several categories [4] such as names of people, places, organisations, items and so on. The essential task in IE and higher level NLP applications is named entity identification. Fine-tuned BERT- based multilingual models for NER in native Indian languages such as Tamil, Malayalam, Marathi, and Bengali are becoming increasingly popular in recent times. These models are pre-trained on massive volumes of multilingual data before being fine-tuned on NER-specific datasets. These models may attain state-of-the-art performance in NER for the different Indian languages by using pre-training and fine-tuning.

The BERT (Bidirectional Encoder Representations from Transformers) architecture has revolutionized NLP in recent years by introducing the concept of pre-training large

language models and fine-tuning them for specific tasks. The multilingual BERT model is capable of processing multiple languages and has been shown to be effective in various NLP tasks. By fine-tuning BERT-based models on NER datasets in Indian languages, it is possible to develop models which can identify and extract named entities from text data automatically in these languages. This can significantly improve the efficiency and accuracy of various NLP applications in these languages.

The Contributions of the proposed model are:

1) Improved NER Performance

- The proposed model has achieved state-of-the-art performance in NER tasks across multiple languages.

2) Multilingual NER

- The Fine-tuned BERT model excels at cross-lingual NER, accurately identifying named entities in diverse languages
- Multilingual NER applications leverage BERT's language understanding, enabling entity extraction and cross-lingual information processing.

3) Named Entity Disambiguation

- The proposed model can help resolve coreference and link entities to unique referents.
- It also improves the quality and coherence of extracted information.

This paper is organized as follows: Section II discusses existing Multilingual models for NER according to the paper's organization. The proposed methodology and architecture are addressed in Section III. The findings and analysis of the suggested approach are detailed in Section IV of the paper, while Section V presents the overall conclusion.

II. RELATED WORKS

Recent advances in deep learning have led to the development of more powerful NER models, such as the Transformer encoder. The paper [5] proposes a new NER architecture that incorporates direction and relative distance aware attention and un-scaled attention to improve the performance of the Transformer in NER. Another paper, [6] proposes a classifier-combination experimental framework for NER that uses four diverse classifiers. The framework achieves a performance of 91.6F without additional training

resources. In the clinical domain, researchers in [7] emphasize the significance of domain-specific information and tools to achieve accurate Named Entity Recognition (NER) in clinical records. They delve into various methods employed for identifying anatomical structures, observations, location qualifiers, and procedures within the clinical context. On a different note, the construction industry is addressed in [8], where the authors tackle the need for an automated specification reviewing model. They propose an NER model with a Bidirectional-LSTM architecture to effectively address this requirement.

In the realm of NER, there have been notable advancements in different domains. One such study introduces a novel approach that combines Latent Dirichlet Allocation and query log data to enhance the accuracy of Named Entity Recognition in Queries (NERQ) [9]. The proposed method demonstrates superior performance compared to existing models in this context. Furthermore, another study presents a hybrid method specifically tailored for precise NER in the medical domain, outperforming previously established models [10]. This approach showcases its potential for advancing NER techniques within the medical field. Additionally, an automated system designed for identifying and linking entities in tweets exhibits improved accuracy and efficiency, contributing to the development of NER capabilities in social media analysis [11].

A graph-based NER model in [12] achieves superior performance in predicting flat and nested named entities. It outperforms previous approaches, highlighting its effectiveness with complex entity structures. [13] addresses NER challenges in code-mixed social media, comparing different ML models. These studies advance NER models for nested entities and code-mixed data. Improving NER in low-resource languages is explored in [14] for Tamil and [15] for Malayalam, emphasizing domain-specific approaches and resource adaptation. Event extraction is a valuable task in natural language processing (NLP), particularly in the context of social media. The paper [16] proposes a two-step event extraction process using a binary classifier and the CRF sequence labeling method. This process can be used to extract events from social media posts, which can then be used to aid in disaster response. The paper highlights the practical applications of event extraction from social media, contributing to the broader field of NLP. Efforts have been made to improve NLP models in domains with limited annotated examples, particularly in the medical text domain, by employing techniques such as pre-training, out-of-vocabulary term optimization, and hyperparameter tuning [17].

In [18], Transformer-based pre-trained models adjusted for NER in Hindi and Bangla demonstrate the effectiveness of the MuRIL Big model. This highlights the potential of pre-trained models in NER tasks. Additionally, in Chinese NER, [19] introduces an improved architecture with enhanced position embedding and self-attention calculations for the Transformer encoder, achieving superior performance on Chinese NER datasets. These studies contribute to domain-specific advancements in NER for specific languages. Content summarization [20] is an important aspect of this research, as it helps condense the information while retaining the essential details.

III. METHODOLOGY

A. Dataset Collection

The WikiAnn dataset serves as a valuable resource for NER research in Indian languages, including Malayalam, Marathi, Tamil, and Bengali. It offers annotated data that can be utilized for various investigations, such as exploring the impact of different factors on NER performance and developing efficient NER models. The dataset enables studying challenges related to multilingual or code-switching text, addressing out-of-vocabulary entities, and evaluating different NER algorithms. It provides insights into the strengths and limitations of rule-based, machine learning-based, and deep learning-based approaches, as well as the effects of engineering strategies on NER performance. Overall, the WikiAnn dataset facilitates comprehensive analysis and advancement in NER for Indian languages.

B. Preprocessing

Preprocessing the wikiAnn dataset is a crucial step in preparing data for NER tasks. It involves cleaning the data by removing unnecessary characters and tags, tokenizing and normalizing the text, performing POS tagging and NER labeling for grammatical functions and named entity identification, and aligning the text with annotated data in different languages. These preprocessing steps ensure high-quality data for training and evaluating NER models on the wikiAnn dataset, enabling multilingual NER research.

C. BERT Architecture

BERT uses Transformer, an algorithm that learns relationships between words (or sub words) in text. In its simplest form, the transformer has two different functions, an encoder receives the input and a decoder generates the prediction function. Because the goal of BERT is to generate linguistic patterns, only one encoder is required.

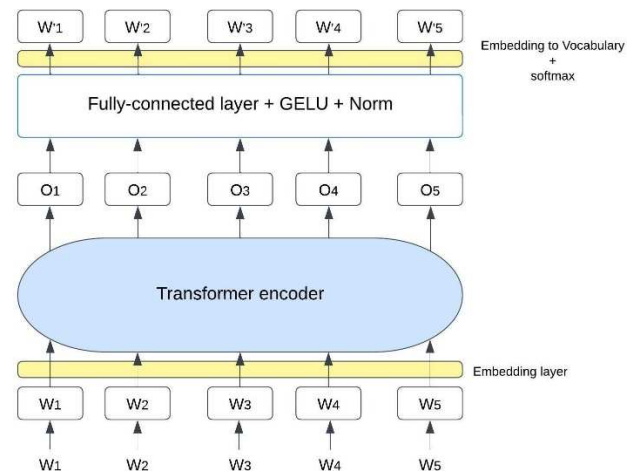


Fig. 1. Bert Model Architecture

The Transformer encoder in Fig.1 is designed to read the entire message one line at a time, unlike standard markers that read text sequentially. It considers both left and right context by utilizing a self-attention mechanism, making it bidirectional in nature. This allows the model to understand the content of a word based on the surrounding context.

The self-attention mechanism involves obtaining query representations (Q), key representations (K), and value representations (V) through linear transformations (XW_Q , XW_K , XW_V). The attention score is computed by applying

a SoftMax function to the dot product of Q and K, scaled by the square root of the dimensionality (d). The output is then calculated by multiplying the attention weights with the value representations (V). They are shown in the equations 1, 2, 3, 4, 5 given below

$$Q = XW_Q \quad (1)$$

$$K = XW_K \quad (2)$$

$$V = XW_V \quad (3)$$

$$\text{Attention score} = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}}\right) \quad (4)$$

$$\text{Output} = \text{Attention} \cdot V \quad (5)$$

On the other hand, the Transformer encoder layer includes a feed-forward neural network that applies a series of linear transformations (W_1 , b_1 , W_2 , b_2) and the GELU activation function. The output of the feed-forward network is added to the input (X), and the result is passed through layer normalization. This process helps capture complex relationships and further refine the representations.

By combining the self-attention mechanism and the feed-forward neural network in the Transformer encoder, the model can effectively capture contextual dependencies and learn meaningful representations for the input sequence.

$$\text{Feed-Forward Network} = \text{GELU}((W_1 + b_1)W_2 + b_2) \quad (6)$$

$$\text{Output} = \text{Layer Norm}(X + \text{Feed-Forward Network}) \quad (7)$$

In the proposed model, we have fine-tuned the encoder by incorporating a task-specific layer. This additional layer enhances the model's ability to capture task-specific information and improve performance. The encoder's output is combined with this task-specific layer, as depicted in equations 6 and 7. To ensure compatibility with the embedding matrix, the output vector is normalized by dividing it by the average size. Subsequently, the SoftMax function is employed to calculate the probability distribution of each word in the dictionary. It is noteworthy that the BERT loss function exclusively considers predictions for masked values, disregarding predictions for unmasked words. This selective approach allows the model to focus solely on the relevant masked words, facilitating more accurate training and optimization.

IV. RESULTS AND DISCUSSION

BERT is a pre-trained language model that is fine-tuned on specific NLP tasks such as NER. It uses a transformer architecture and is trained for 200 epochs on a large corpus of text data. Fine-tuning the pre-trained BERT model on NER task involves training the model on labeled data to recognize and classify entities in text such as people, organizations, and locations.

We used evaluation metrics such as precision, recall, f1 and accuracy to evaluate the performance of the trained model.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (8)$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (9)$$

In equation 8, Precision is the fraction of instances that are accurately classified as positive by the model. In equation 9, Recall is the fraction of positive instances properly recognized in the data.

Accuracy measures the proportion of correct predictions from all events, including both true positives and negatives. These measures are calculated from the number of true positives, negatives, negatives and negatives and form the basis for performance evaluation.

A. Location (LOC) metrics

TABLE I. COMPARISON OF DIFFERENT PERFORMANCE METRICS OF LOCATION ENTITY

Language	Precision	Recall	F1
Tamil	83.17%	86.95%	82.65%
Malayalam	79.8%	91.26%	85.50%
Marathi	88.3%	91.56%	87.15%
Bengali	94.2%	94.82%	90.38%

The table I presents the language classification metrics for the studied languages and for the LOCATION entity. The precision values range from 79.8% for Malayalam to 94.2% for Bengali. The high precision achieved for Bengali demonstrates the model's ability to accurately identify texts in this language. Similarly, recall values range from 86.95% for Tamil to 94.82% for Bengali, indicating the proportion of correctly classified instances identified by the model.

B. Organisation (ORG) metrics

TABLE II. COMPARISON OF DIFFERENT PERFORMANCE METRICS OF ORGANISATION ENTITY

Language	Precision	Recall	F1
Tamil	82.5%	82.1%	82.2%
Malayalam	81.3%	76.5%	78.8%
Marathi	84%	85.6%	88.8%
Bengali	95.4%	95.5%	95.4%

The obtained results are presented in the table II, which summarizes the precision, recall, and F1 scores for the evaluated languages. The precision scores for Tamil, Malayalam, Marathi, and Bengali were measured at 82.5%, 81.3%, 84%, and 95.4% respectively. These values indicate the system's ability to provide accurate results with minimal false positives. Similarly, the recall scores for the languages stood at 82.1%, 76.5%, 85.6%, and 95.5% respectively, indicating the system's capability to retrieve relevant information effectively.

The F1 scores, which represent the harmonic mean of precision and recall, were found to be 82.2%, 78.8%, 88.8%, and 95.4% for Tamil, Malayalam, Marathi, and Bengali. These scores demonstrate the balanced performance of the system in terms of precision and recall for each language.

C. Person (PER) metrics

TABLE III. COMPARISON OF DIFFERENT PERFORMANCE METRICS OF PERSON ENTITY

Language	Precision	Recall	F1
----------	-----------	--------	----

Tamil	86.3%	87.9%	87.15%
Malayalam	86.2%	84.4%	85.3%
Marathi	92.08%	93.6%	92.8%
Bengali	96.6%	93.5%	95.04%

The results in the table III presents the precision, recall, and F1 scores for person entity recognition in Tamil, Malayalam, Marathi, and Bengali. Precision scores range from 86.3% to 96.6%, indicating accurate identification of person entities. Recall scores range from 84.4% to 93.6%, reflecting effective retrieval of person entities. F1 scores range from 85.3% to 95.04%, demonstrating a balanced performance in precision and recall. Bengali achieves the highest scores, indicating superior performance, while Malayalam shows slightly lower scores. These results highlight the importance of considering language-specific characteristics in person entity recognition systems.

D. Overall Metrics

TABLE IV. COMPARISON OF DIFFERENT PERFORMANCE METRICS

Language	Precision	Recall	F1	Accuracy
Tamil	84.1%	85.7%	85%	92.2%
Malayalam	82.1%	82.1%	82.2%	91.7%
Marathi	88.3%	89.2%	88.7%	94.2%
Bengali	95.3%	94.8%	95.1%	96.0%

This table IV provides an overview of the overall performance of the language processing system for Tamil, Malayalam, Marathi, and Bengali. The precision scores range from 82.1% to 95.3%, indicating accurate results with low false positives. Recall scores range from 82.1% to 94.8%, demonstrating effective retrieval of relevant information. F1 scores range from 91.7% to 96.0%, representing a balanced performance in precision and recall for each language. These results highlight the system's ability to deliver accurate and relevant results across the evaluated languages.

E. Precision

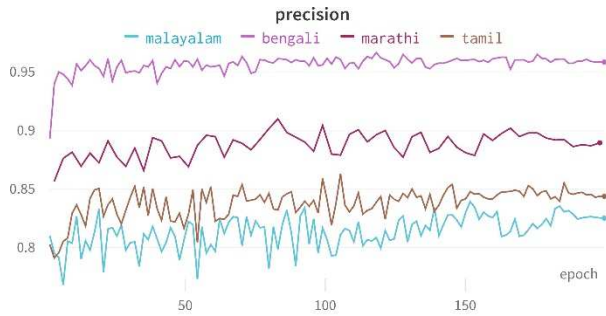


Fig. 2. Precision plot for Tamil, Malayalam, Marathi and Bengali

The precision plot for the four languages (Tamil, Malayalam, Marathi, and Bengali) shown in the Fig.2 describes varying levels of accuracy in identifying named entities. The precision values indicate that the NER system performs moderately in Tamil and Malayalam, with precision rates of 84.1% and 82.1% respectively. In Marathi, the system achieves higher precision at 88.3%, while in Bengali, it demonstrates exceptional accuracy with a precision rate of 95.3%. These results suggest that the NER system performs differently across languages, achieving higher precision in Marathi and Bengali compared to Tamil and Malayalam.

F. Recall

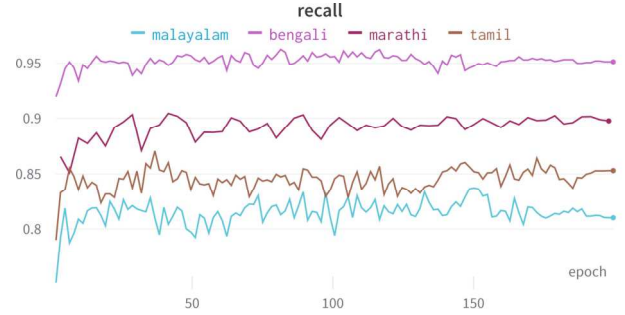


Fig. 3. Recall plot for Tamil, Malayalam, Marathi and Bengali

As shown in the Fig.3 the recall values for four languages (Tamil, Malayalam, Marathi, and Bengali) in the NER system are relatively high, indicating its ability to correctly identify named entities in the text. The performance is consistent across languages, with values ranging from 82.1% to 94.8%. Bengali achieves the highest recall, while Malayalam shows a slightly lower performance. Overall, the NER system demonstrates effective recognition of named entities, with some variation in specific languages.

G. F1-score

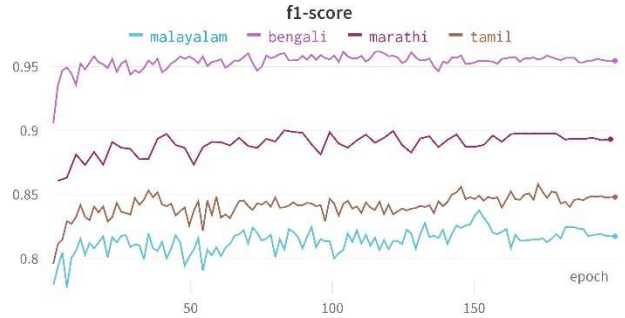


Fig. 4. F1 score plot for Tamil, Malayalam, Marathi and Bengali

The F1 plot presents the overall language identification performance for different languages which is shown in Fig.4. The F1 scores reveal variations in accuracy and completeness across the languages. Bengali achieves the highest F1 score of 95.1%, followed by Marathi with 88.7%. Tamil and Malayalam show comparatively lower F1 scores of 85% and 82.2% respectively. The F1 plot emphasizes the need to consider language-specific challenges and characteristics in language identification tasks. It provides valuable insights for evaluating the effectiveness of the language identification system and identifying areas for improvement.

H. Accuracy

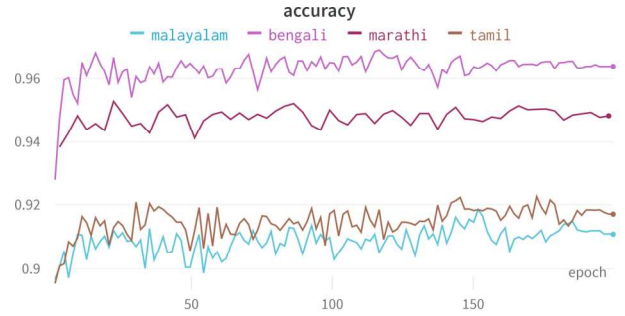


Fig. 5. Accuracy plot for Tamil, Malayalam, Marathi and Bengali

As shown in the Fig.5, from the accuracy plot, it can be inferred that the language identification system achieved high accuracy rates for the languages Tamil, Malayalam, Marathi, and Bengali. The accuracy values for these languages are 92.2%, 91.7%, 94.2%, and 96% respectively. This suggests that the system was able to correctly identify the languages with a high level of accuracy, demonstrating its effectiveness in language recognition tasks. The higher accuracy values indicate a strong performance of the system in accurately classifying the input texts into their respective languages.

I. Quantitative Analysis

TABLE V. COMPARISON OF F1 SCORES OF PROPOSED MODEL AND INDICNER MODEL

Model	Language	LOC	ORG	PER	Overall
IndicNER [21]	Tamil	53.06	93.33	82.67	77.84
	Malayalam	61.59	87.79	84.49	81.67
	Marathi	66.33	88.66	83.70	81.88
	Bengali	73.73	84.29	77.74	80.20
Proposed model	Tamil	82.65	82.2	87.15	85
	Malayalam	85.5	78.8	85.3	82.2
	Marathi	87.15	88.8	92.8	88.7
	Bengali	90.38	95.4	95.04	95.1

As shown in the table V, the proposed model for Named Entity Recognition outperformed the IndicNER model when trained on Tamil, Malayalam, Marathi and Bengali languages using the WikiAnn dataset. We evaluated the performance of both models using the F1 score for three entity tags: LOC, ORG, PER as well as the overall F1 score. The proposed model achieved an overall F1 score of 85%, 82.2%, 88.7% and 95.1% when trained on Tamil, Malayalam, Marathi and Bengali Respectively. which is significantly higher compared to other model's overall F1 score of 77.84%, 81.67%, 81.88% and 80.20% when trained on Tamil, Malayalam, Marathi and Bengali Respectively.

These results indicate that the proposed model for NER demonstrates superior performance across all evaluated languages. The overall F1 scores achieved by the proposed model consistently outperformed the IndicNER model. This suggests that the proposed model is more effective in recognizing named entities, resulting in improved accuracy and precision in identifying LOC, ORG, and PER tags.

J. Qualitative Analysis

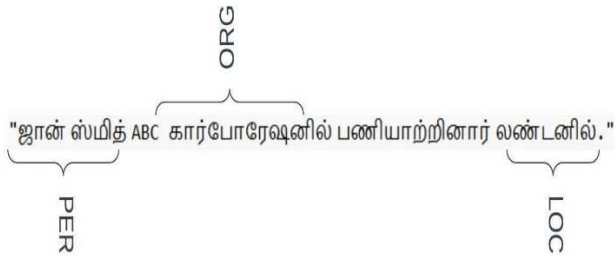


Fig. 6. Predicted tags for the text in tamil

The qualitative analysis of the proposed methodology's output unveils valuable insights regarding the entities identified in the text. The analysis showcases the potential of BERT, a cutting-edge language model, in accurately recognizing entities such as PERSON, ORGANIZATION, and LOCATION. The NER model demonstrates proficiency in

recognizing person names, as it successfully identifies person entities in the given Tamil text. The model shows accuracy in recognizing organization names, as it correctly identifies the organization entity as shown in fig.6. Additionally, the model displays proficiency in identifying locations, as it accurately identifies the location entity. The analysis uncovers that Jan Smith, an individual whose gender and specific attributes remain undisclosed, has a connection with ABC Corporation, an organization based in London. These findings exemplify the capability of language models, such as BERT, in entity recognition tasks.

V. CONCLUSION

The fine-tuned BERT-based multilingual Named Entity Recognition (NER) model, trained on the WikiAnn dataset for Malayalam, Marathi, Tamil, and Bengali languages, demonstrates impressive performance. Bengali achieved the highest metrics with precision of 95.3%, recall of 94.8%, and accuracy of 96.0%. This model effectively identifies named entities in text data, holding significant potential for Natural Language Processing (NLP) applications such as information extraction, entity linking, and named entity disambiguation in these languages. Accurate recognition of named entities contributes to improved understanding and analysis of text data. Limitations include language coverage constraints, data availability dependency, named entity variations, and domain-specific challenges. These factors require cautious deployment of the model in practical applications. The future scope of mBERT involves expanding its application across diverse industries and domains, leveraging its multilingual capabilities and task-oriented fine-tuning to advance various NLP tasks, including information extraction, document classification, sentiment analysis, machine translation, question answering, and knowledge graph construction in multilingual settings.

REFERENCES

- [1] H. Shelar, G. Kaur, N. Heda, and P. Agrawal, "Named entity recognition approaches and their comparison for custom ner model," Science Technology Libraries, 2020.
- [2] B. M. Gurusamy and R. Kumar, "Survey of text document summarization based on ensemble topic vector clustering model," pp. 831–847, 10 2022.
- [3] G. B. Mohan and R. P. Kumar, "A comprehensive survey on topic modeling in text summarization," pp. 231–240, 2022.
- [4] Y. Wang, Y. Sun, M. Zuchang, L. Gao, and Y. Xu, "Named entity recognition in chinese medical literature using pretraining models," Scientific Programming, vol. 2020, pp. 1–9, 09 2020.
- [5] H. Yan, B. Deng, X. Li, and X. Qiu, "Tener: Adapting transformer encoder for named entity recognition," 2019.
- [6] R. Florian, A. Ittycheriah, H. Jing, and T. Zhang, "Named entity recognition through classifier combination," Pro- ceedings of CoNLL-2003, 2004.
- [7] S. R. Kundeti, J. Vijayananda, S. Muggiga, and M. Kalyan, "Clinical named entity recognition: Chal- lenges and opportunities," 2016.
- [8] F. Hassan, T. Le, and X. Lv, "Addressing legal and con- tractual matters in construction using natural language processing: A critical review," Journal of Construction Engineering and Management, 2021.
- [9] J. Guo, G. Xu, X. Cheng, and H. Li, "Named entity recognition in query."
- [10] R. Ramachandran and K. Arutchelvan, "Named entity recognition on bio-medical literature documents using hybrid-based approach," Journal of Ambient Intelligence and Humanized Computing, 03 2021.
- [11] S. K. P. Barathi Ganesh H B, "Amrita - cen@neel: Identification and linking of twitter entities," 05 2015.
- [12] J. Yu, B. Bohnet, and M. Poesio, "Named entity recognition as dependency parsing," 05 2020.

- [13] S. S. and M. Shrivastava, ““kanglish alli names!” named entity recognition for kannada-english code-mixed social media data,” 10 2022.
- [14] R. Murugathas and U. Thayasivam, “Domain specific named entity recognition in tamil,” pp. 1–6, 07 2022.
- [15] S. P S and A. Pillai, “Towards an efficient malayalam named entity recognizer analysis on the challenges,” *Procedia Computer Science*, vol. 171, pp. 2541–2546, 01 2020.
- [16] S. Burramsetty, N. Gonugunta, S. Uppu, and S. Nunna, “Event extraction from telugu-english code mixed social media text,” pp. 946–951, 06 2022.
- [17] A. Kormilitzin, M. Hofer, P. Goldberg, and A. Nevado- Holgado, “Few-shot learning for named entity recogni- tion in medical text,” 2018.
- [18] S. Singh, P. Jawale, and U. Tiwary, “Transformer based ner models for hindi and bangla languages,” pp. 1536– 1542, 01 2022.
- [19] M. Sun, L. Wang, T. Sheng, Z. He, and Y. Huang, “Chinese named entity recognition using the improved transformer encoder and the lexicon adapter,” pp. 197– 208, 09 2022.
- [20] B. M. Gurusamy and R. Kumar, “Lattice abstraction- based content summarization using baseline abstractive lexical chaining progress,” *International Journal of In- formation Technology*, vol. 15, 10 2022.
- [21] A. Mhaske, H. Kedia, S. Doddapaneni, M. M. Khapra, P. Kumar, R. Murthy V, and A. Kunchukuttan, “Naama- padam: A large-scale named entity annotated data for in- dic languages,” *arXiv preprint arXiv:2212.10168*, 2022.